**Foundation for Virtual and Augmented Reality Systems**

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**Week – 02**

**Lecture – 10**

## Lec 10: Cybersickness

Hello and welcome to the NPTEL MOOC's course on Foundations for Virtual and Augmented Reality Systems. We are going to start our lecture number 6 on Cybersickness. Before we start, let us quickly recap what we have learned so far. We talked about the basic concepts, and we had gone through the definitions of those basic concepts. VR, AR and MR. We have also talked about applications and the categories of systems that are there.

In the previous lecture, we talked about the issues and challenges that are of major concern in virtual and augmented reality systems. Also in between, we talked about in brief the historical evolution of these fields. So, in the previous lecture we talked about a few concerns, but one more concern we are going to talk about in detail in this lecture that is cybersickness which is primarily relevant for virtual reality systems.

In particular, we are going to talk about three aspects of cybersickness namely causes, what causes cybersickness, then how to measure or evaluate cybersickness and what are the mitigation strategies that we can adopt. Now, cybersickness is known by another name that is Visually Induced Motion Sickness or VIMS in short. So, in the literature you may come across both the terms cybersickness or CS or VIMS which stands for visually induced motion sickness. So, this is a condition that can affect individuals when they interact with virtual reality technology.

A condition that affects individuals when they interact with virtual reality technology. Note that we are primarily focusing on virtual reality technology. And CS or VIMS is primarily relevant in the domain of virtual reality. Now, how do we know that CS has happened or is going to happen? What are the symptoms?

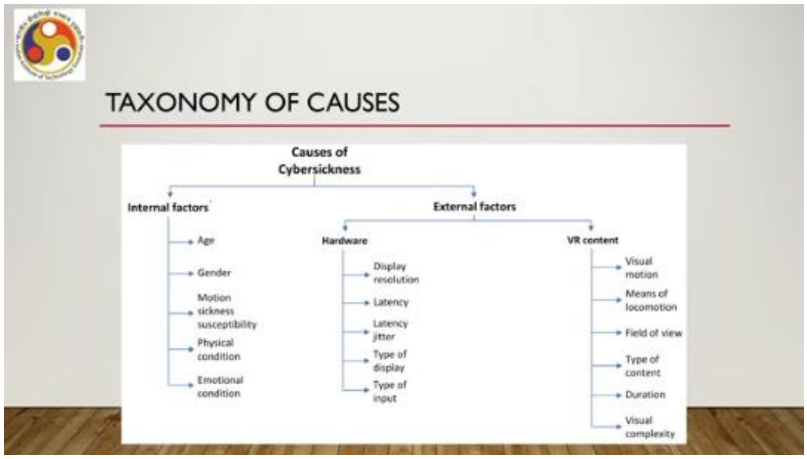
There are, in fact, a large number of symptoms that may indicate the onset of CS or the CS itself. Some are well known, some are not so well known. For example, nausea, headache, excessive sweating, dryness of mouth or even fullness of stomach, vomiting or tendency, vomiting tendency particularly, dizziness, disorientation, fatigue, strange hand-eye coordination. So, typically the coordination happens in not the expected way.

There are many more such conditions, but these are such symptoms but these are common ones. So, if one or more of these symptoms are there during VR experience, then we can assume that cyber sickness is happening or going to happen. Now, it can be caused by many factors. So, not a single factor is responsible.

There can be many factors that can cause cyber sickness. Finding a feasible solution to reduce the impact of CS is extremely important in the context of VR systems. It greatly enhances the overall user experience, which in turn makes VR applications more appealing to a wider range of people. So, clearly from the usability and acceptability point of view, reduction of cyber sickness is very important.

So, in order to understand cyber sickness, let us focus on three aspects of it. First one is causes: what causes cyber sickness? Then the second one is evaluation: how do we measure cyber sickness? And the third one is mitigation: how do we overcome the challenges associated with cyber sickness? So, let us start with causes.

Here I show a basically a summary of different factors that can cause cyber sickness. There are broadly two factors, internal and external. Now within external, there can be two sub-factors: hardware and VR content. We will go through each of these broad categories as well as the subcategories in detail in subsequent lectures.



Let us start with the internal factors. These are a set of individual characteristics and conditions that can contribute to an increased likelihood of experiencing cyber sickness. What it tells is that there are factors that are individual characteristics and conditions. In other words, these are inherent to users and can include a variety of different elements. Since these are inherent to users, we are calling them internal factors.

There can be many such internal factors as we have seen in the summary. So, one of those is age. Studies have indicated there is a relationship between age and CS or cyber sickness susceptibility. It has been found through studies that children between 2 to 12 years of age may exhibit increased sensitivity to cyber sickness. In other words, they are more likely to fall prey to cyber sickness, but that tendency decreases

as those children grow older. On the other hand, older persons are likely to experience higher levels of CS. But CS is generally not there or uncommon among young children. So, those between say 12 and 30. 12 and 20, 12 and 25.

So, during this, in this age group, it is highly unlikely that users will feel cyber sickness. A study also revealed that CS can be independent of age when the user is viewing general VR content, but age may be a factor for controllable content. So, in the case of general VR content, the relationship between age and CS is not so clear, but in the controllable content, the relationship is more visible. The next internal factor is gender.

The impact of gender on cyber sickness has been the subject of numerous studies. However, there is no conclusive result. Some studies have found that women users are more likely to have more severe cases of CS. But other studies revealed there are not any discernible differences between male and female participants in terms of the presence and severity of cyber sickness. So, although some studies have found that it is likely to more affect women users, other studies have revealed that there is no such relationship.

It has also been found that females using head-mounted displays, HMDs, are associated with an increase in CS, an increase in cyber sickness. So, those female users who are using head-mounted displays are likely to experience more cyber sickness. One of the possible reasons may be the inter-pupillary distance or IPD. of the user. Now, this distance determines how we are going to have stereoscopic vision.

What is the experience of having stereoscopic vision through the HMDs? So, a greater proportion of females having trouble finding a good fit indicates that they are likely to experience more CS as compared to males. So, what it implies is that we need to have VR

displays with a wider IPD adjustable range catering to female users. But overall, we cannot say that there is a definite relationship between cyber sickness and gender. Studies have reported contradictory results.

An internal factor we can talk about is motion sickness susceptibility. So, motion sickness we can refer to in short as MS. So, MS susceptibility is another internal factor. Now, research has explored connections between CS and MS. What is MS?

It occurs when the brain receives conflicting signals from various sensory organs. causing symptoms like nausea and dizziness similar to cyber sickness. VR technology can produce realistic movements, that is the goal of VR, and visuals that are again part of a realistic experience. So, the user feels that he or she is in a real world environment. Accordingly, the user may experience similar sensations to real world motion because the VR world is recreating the real world.

So, those having MS susceptibility, because they are feeling a real world environment in the VR system, they tend to have CS susceptibility as well. Several studies have indicated a positive correlation between MS susceptibility and CS risk. So, individuals with higher MS susceptibility are likely to experience more severe CS symptoms; that is the implication of the positive correlation. However, that is not necessarily always true. Prior experience with VR

systems may reduce the effect of MS susceptibility on CS. So, even if a user has MS susceptibility, but if the user is exposed to VR systems beforehand, then in subsequent usage the user may not feel severe CS. Additionally, repeated exposure to VR has also shown to be effective in reducing CS symptoms over time. So, Again in this case, even if the user is susceptible to MS or motion sickness, still if the user is exposed to VR experience over time,

repeated exposure, then the severity of CS in future use may be less, although the user is susceptible to MS. So, the positive correlation does not hold. If either of these conditions is satisfied, or both of these conditions are satisfied, that is prior exposure and repeated exposure. Another internal factor can be physical conditions. One of the physical conditions that is responsible for CS is body awareness.

That is, the ability of the user to attend to and perceive bodily sensations. Now, that awareness gets affected by poor sleep quality. So, poor sleep quality may lead to or contribute to CS, or increase the severity of CS. People with greater retinal eccentricity

may also be more susceptible to CS. Individuals with neurological conditions such as multiple sclerosis may experience more severe CS symptoms.

Besides specific physical conditions, the physical position of the user can also impact susceptibility to CS. So, when the user is using the VR system in a standing posture, then it is likely to have more induction of CS as compared to if the user is sitting and using the same system. So, when the user is standing, the likelihood of CS or severity of CS is likely to be more as compared to if the user is sitting. So, the position also affects

the onset of CS. Another internal factor can be emotions or emotional conditions at the time of usage. So, negative emotions such as anxiety, stress, and frustration are typically associated with an increased likelihood of experiencing visually induced motion sickness or cyber sickness, CS. Personality traits such as neuroticism may also increase the likelihood of experiencing CS.

So, these are the internal factors, factors that are internal to the user: age, gender, motion sickness susceptibility, physical condition, and emotional condition. Then there are external factors, factors that are not subjective to users' mental or physical conditions. So, one category of external factors is hardware. So, let us have a look at this external factor first. Initially, in the early days of virtual reality, people tended to associate CAs with

inferior quality hardware. They thought that with the advancement of technology, these problems would go away. However, even though the technology has advanced, many of those problems still remain. That indicates that problems are not related to inferior hardware quality. There is something more fundamental there.

So, some aspects of devices, no matter how advanced they are, are responsible for inducing CS. So, one of those factors is display resolution. Studies show that higher resolution leads to better experience and reduced CS up to a certain point. It is not necessary that it will keep on reducing.

But better resolution can definitely reduce CS, but up to a certain point. A study evaluated CS at resolutions 1K, 2K, 3K, and 4K. So, different resolutions were compared to see the effect of resolution on CS. So, what was the finding? There was a substantial difference.

In the induction of CS when the resolution changed from 1K to 2K. So, when the resolution changed to 2K from 1K, CS induction reduced. However, that reduction was not significant when resolutions were further changed from 2K to 3K and 3K to 4K. That means up to a change of 2K, it is likely to affect the induction of CS.

But if we change the resolution beyond that, if we increase the resolution, then it is not likely to change anything related to induction of CS. Another hardware-related factor is latency. Latency is the time delay between a user's movement and the corresponding change in the virtual world. So, you are doing something in the physical world which gets reflected in the virtual world. The delay between these two, the time gap between these two, is called latency.

Now, several studies have investigated the relationship between CS and latency. In the context of HMD-based VR experience, HMD or head-mounted display-based VR experience, it is found that sickness increases with latency. The more latency there is, the more likely it is to induce sickness. A related external factor. Another hardware-related external factor is latency jitter.

So, latency is the delay between what you do in the physical world and what is happening in the virtual world due to that action. Jitter is when this time gap keeps on varying. So, variability in latency. So, sometimes you do something in the real world which gets reflected, say, after 1 second in the virtual world. At other times, that same action gets reflected in the virtual world after, say, 2 seconds, although other things remain the same.

So, this is definitely due to the quality of hardware, the components of hardware. So, one is latency, but jitter is variability in latency. Now, this causes irregularities in the timing of the delivery of data. Now, this delay in response can lead to inconsistencies in the experience, causing discomfort and disorientation for the user, which leads to CS. Type of display is another hardware factor which is part of the external factors.

Severity of symptoms. Due to CS can be significantly influenced by the type of display. Earlier we have seen that resolution affects symptoms of CS. Now the type of display also affects the same. A study found that users with HMDs felt more cyber sick as compared to those using conventional monitors.

So if you are using non-immersive VR system, vis-a-vis HMD-based semi-immersive VR system, then the latter category is likely to result in more CS as compared to the previous one. Another study compared quality of experience of stereoscopic 3D videos on different projection devices which comprised flat screens, panoramic screens and VR headsets. And as before, it has been found that users with VR headsets experienced more CS.

So, if you are using or if your users are using HMD-based displays as compared to other traditional displays, they are likely to experience more cyber sickness. Another interesting

finding was that users with higher-end HMDs experienced less cyber sickness as compared to those with lower-end HMDs. These are related, of course, to display qualities in terms of resolution, brightness, clarity, contrast. So, those also affect cyber sickness. Type of input, another important factor or cause for cyber sickness.

Several input methods are used in VR such as hand controllers, treadmills and motion sensors. And depending on the input method, input device, input technology, cyber sickness can vary. Motion controllers such as steering with handlebars may lead to higher CS as compared to non-motion controls such as a joystick. Use of a gamepad controller causes less cyber sickness than using a hand controller. So, these are some of the findings related to types of input and how they affect cyber sickness.

In summary, what we can say is that in this part of the lecture, we have covered some of the causes. So, broadly there are two factors, internal and external. Among internal, there are several factors like age, gender, etc. The external factors can broadly be divided into hardware and content. Hardware we have seen.

Several factors: display type, latency, latency jitter, resolution, input technology, etc. So, we will continue our discussion on these causes and we will cover the remaining causes part that is the VR content which comes under the external factors in the subsequent part of the lecture. So, looking forward to seeing you in the next part of the lecture. Thank you, goodbye, happy learning.